

Design and Implementation of a Microstrip Patch Antenna for 2.4 GHz ISM Band Applications

1. Abstract

This project presents the design and implementation of a compact microstrip patch antenna for Industrial, Scientific, and Medical (ISM) band applications. The objective of the work is to develop a low-profile antenna operating near the 2.4 GHz ISM band for wireless communication systems. The antenna was designed on an FR-4 substrate with overall dimensions of 60 mm $\times$ 60 mm $\times$ 1.6 mm and optimized using CST Studio Suite. A modified rectangular patch structure with an inset-fed microstrip line and edge notches was employed to improve impedance matching and resonant performance.

The antenna performance was investigated through parametric optimization and electromagnetic simulation. The optimized design achieved a resonant frequency of 2.41 GHz with a minimum reflection coefficient (S_{11}) of -18.64 dB, indicating good impedance matching. Furthermore, the antenna exhibited a realized gain of approximately 1.86 dBi at the operating frequency. Radiation pattern analysis demonstrated stable directional characteristics suitable for ISM band applications.

The obtained results confirm that the proposed antenna is capable of operating effectively in the 2.4 GHz ISM band and can be utilized in short-range wireless communication systems such as Wi-Fi, IoT devices, and sensor networks.

2. Objectives

- To design a compact microstrip patch antenna operating in the 2.4 GHz ISM band.
- To analyze the effect of different geometrical parameters on antenna performance.
- To optimize the antenna dimensions for improved impedance matching and radiation characteristics.
- To evaluate the antenna performance in terms of reflection coefficient (S_{11}), gain, efficiency, and radiation pattern.
- To fabricate and experimentally validate the proposed antenna using a Vector Network Analyzer (VNA).

3. Antenna Design

A microstrip patch antenna was designed on an FR-4 substrate having a dielectric constant of 4.3 and a thickness of 1.6 mm. The overall dimensions of the substrate are 60 mm $\times$ 60 mm. The antenna consists of a rectangular radiating patch, an inset-fed microstrip feed line, and a ground plane on the opposite side of the substrate.

To improve the impedance matching and resonant performance, two rectangular notches were introduced at the upper portion of the patch. The feed line was designed to provide approximately 50 Ω input impedance. The antenna geometry was modeled and simulated using CST Studio Suite.

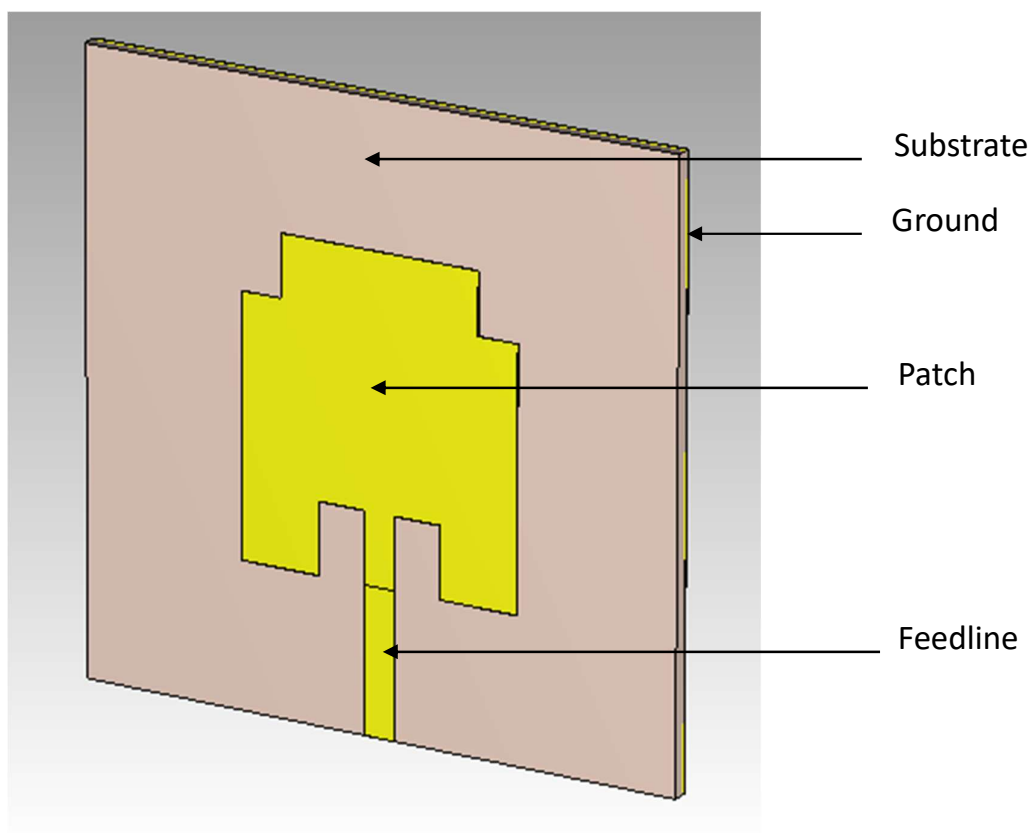


Fig.1: Geometry of the proposed antenna.

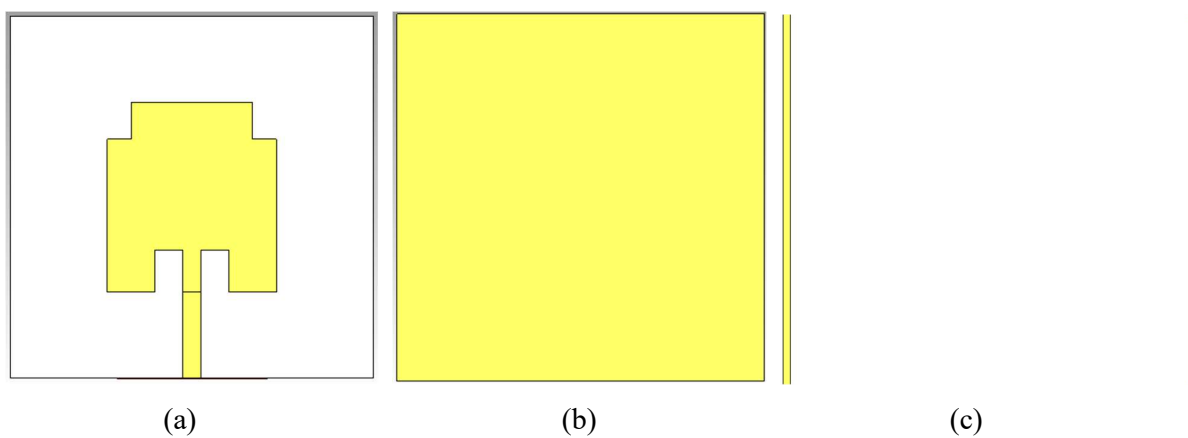


Fig.2: (a) Top view, (b) bottom view and (c) side view of the proposed antenna.

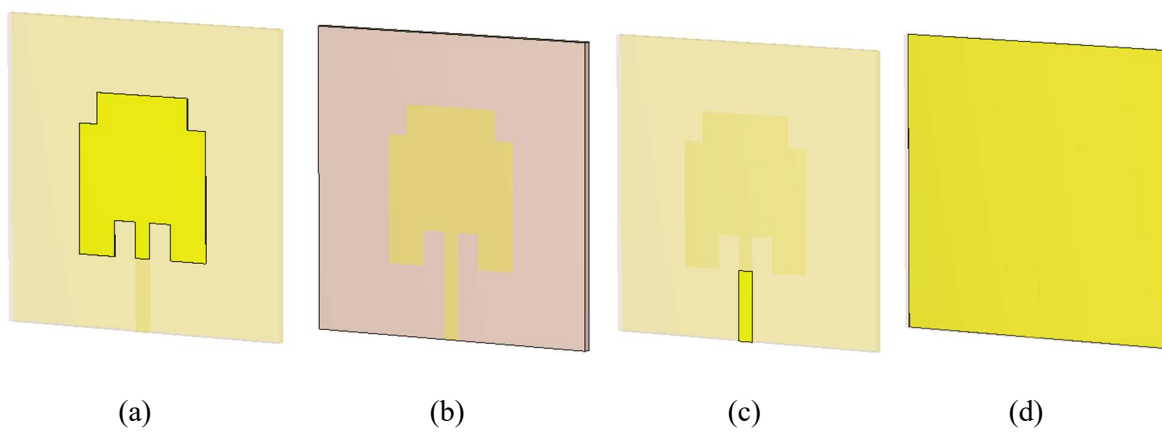


Fig.3. (a) Patch, (b) Substrate, (c) Feedline and (d) Ground

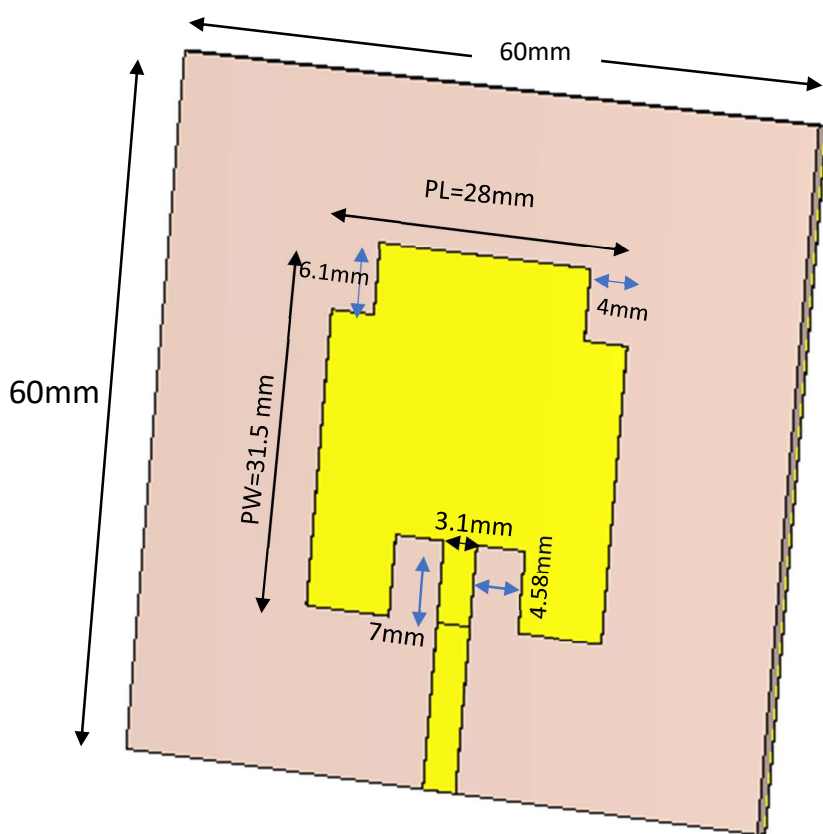


Fig.4. Geometrical dimensions.

Table 1: Geometrical dimensions

PARAMETER	DESCRIPTION	VALUE (MM)
SL	Substrate Length	60
SW	Substrate Width	60
PL	Patch Length	28
PW	Patch Width	31.5
FW	Feed Line Width	3.1
NSW	Top Notch Width	4
NSL	Top Notch Depth	6.1
ST	Feed Side Slot Width	4.58
SLOT_L	Feed Side Slot Depth	7
SH	Substrate Thickness	1.6
GH	Copper Thickness	0.035

4. Effect of Geometrical Variations on the Antenna Performance

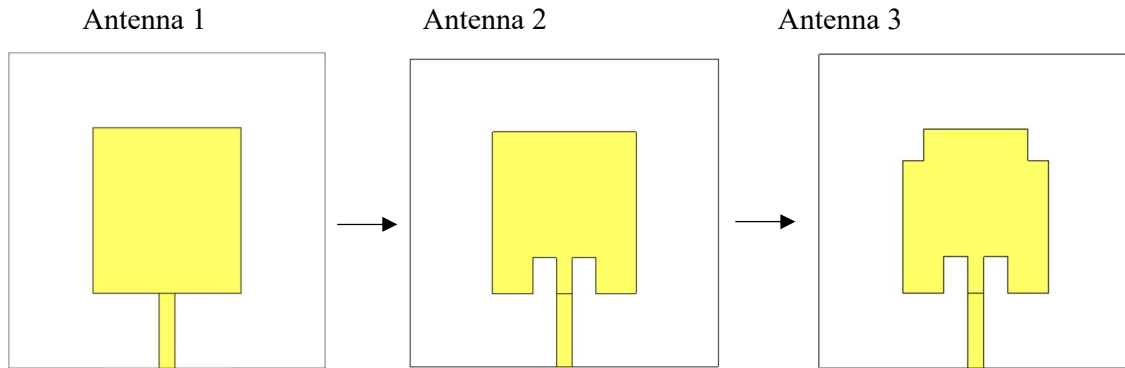


Fig.5: Evolution process of the proposed antenna.

Fig.5 illustrates the evolution process of the proposed antenna. Antenna 1 represents the basic rectangular patch structure. Antenna 2 introduces modifications in the patch geometry to improve impedance matching. Finally, Antenna 3 incorporates additional slots and notches, resulting in improved return loss characteristics and resonance near the desired 2.4 GHz ISM band.

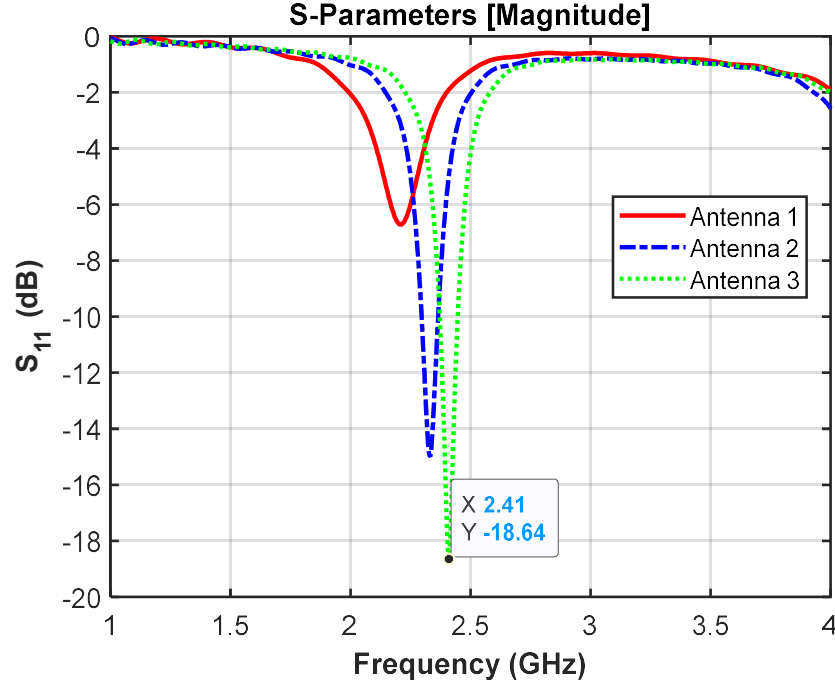


Fig.6: S parameters of the proposed antenna.

Fig.6 shows the reflection coefficient (S_{11}) characteristics of different antenna configurations during the design evolution process. It can be observed that the resonant frequency and impedance matching change significantly with each modification of the antenna geometry. The introduction of slots and notches improves the current distribution on the patch and enhances impedance matching. Among all configurations, the final antenna exhibits the best return loss performance near the desired 2.4 GHz ISM band.

5. Results and Discussions

5.1. Simulated Results

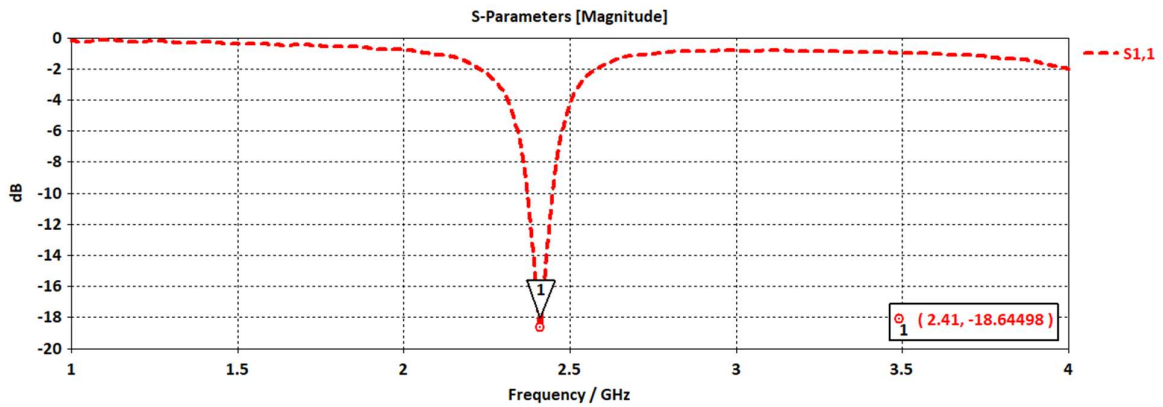


Fig.7: Simulated reflection coefficients (S_{11}) of the proposed antenna.

Fig.7 shows the simulated reflection coefficient of the proposed antenna. The antenna resonates at approximately 2.41 GHz with a minimum S_{11} of -18.64 dB, indicating good impedance matching and efficient power transfer.

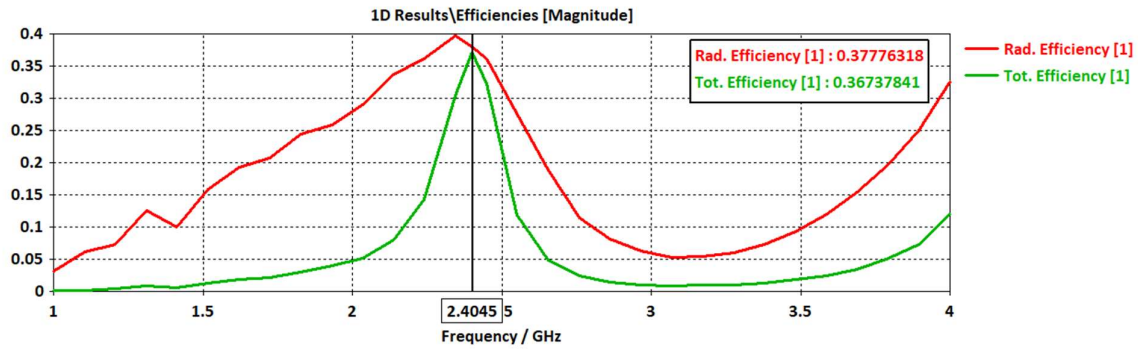


Fig.8: Efficiency of the proposed antenna.

Fig.8 illustrates the radiation efficiency of the proposed antenna as a function of frequency. The efficiency remains relatively high around the operating frequency, indicating low conductor and dielectric losses. High efficiency ensures that a large portion of the supplied power is converted into electromagnetic radiation, thereby improving overall antenna performance.

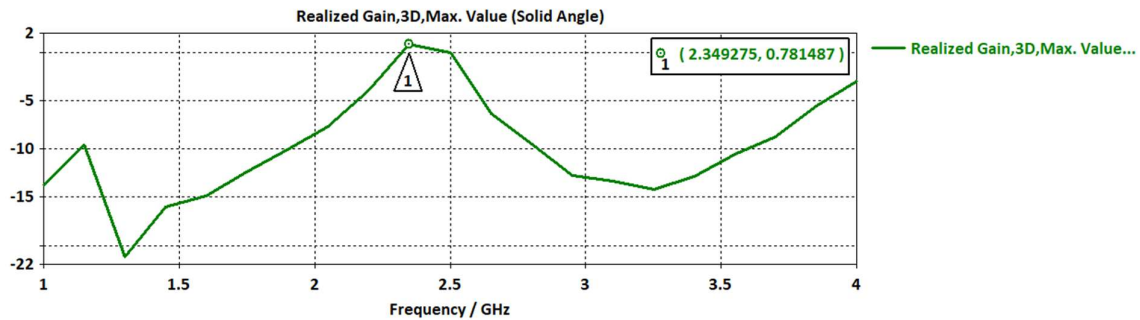


Fig.9: Variation of Gain w.r.t. frequency of the proposed antenna.

Figure 9 shows the variation of realized gain over the operating frequency range. The antenna achieves a maximum gain of approximately 1.86 dBi near the resonant frequency of 2.41 GHz. The gain behavior confirms that the antenna provides adequate radiation performance for short-range wireless communication systems operating in the ISM band.

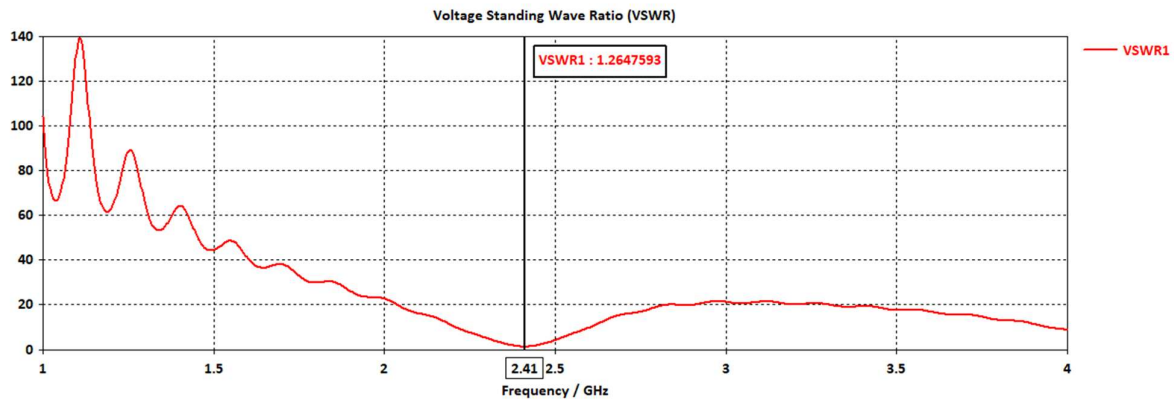


Fig.10: Variation of VSWR w.r.t. frequency of the proposed antenna

Fig.10 presents the Voltage Standing Wave Ratio (VSWR) characteristics of the proposed antenna. At the resonant frequency, the VSWR approaches unity and remains below 2, indicating good impedance matching between the antenna and the transmission line. A lower VSWR value minimizes reflected power and improves power transfer efficiency.

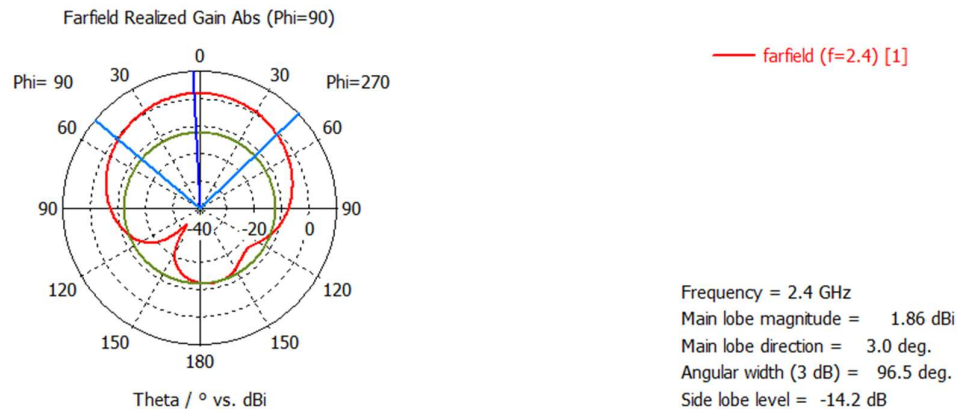


Fig.11: 1D radiation pattern at 2.4 GHz in H-plane

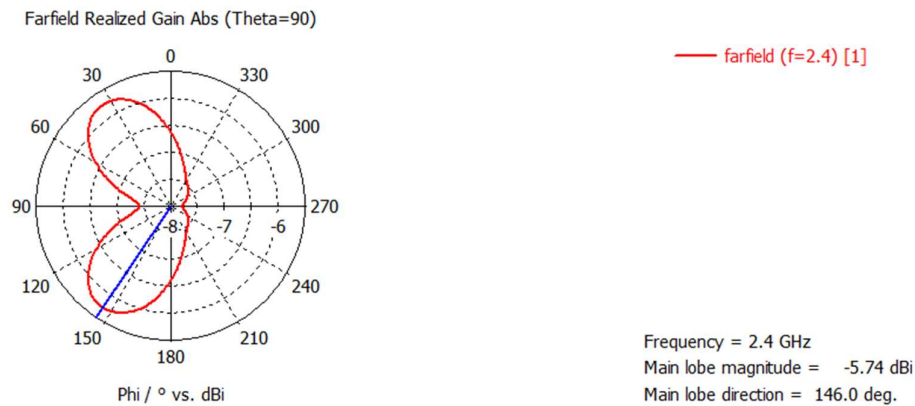


Fig.12: 1D radiation pattern at 2.4 GHz in E-plane

Fig.11 and fig.12 show the H-plane and E-plane radiation patterns of the antenna at 2.4 GHz. Both radiation patterns exhibit a clear main lobe and stable radiation characteristics. The obtained patterns indicate that the antenna radiates effectively at the operating frequency and is suitable for ISM band wireless communication applications.

5.2. Experimental Results



(a)



(b)

Fig.13: (a) Front view and (b) Back view of the fabricated antenna

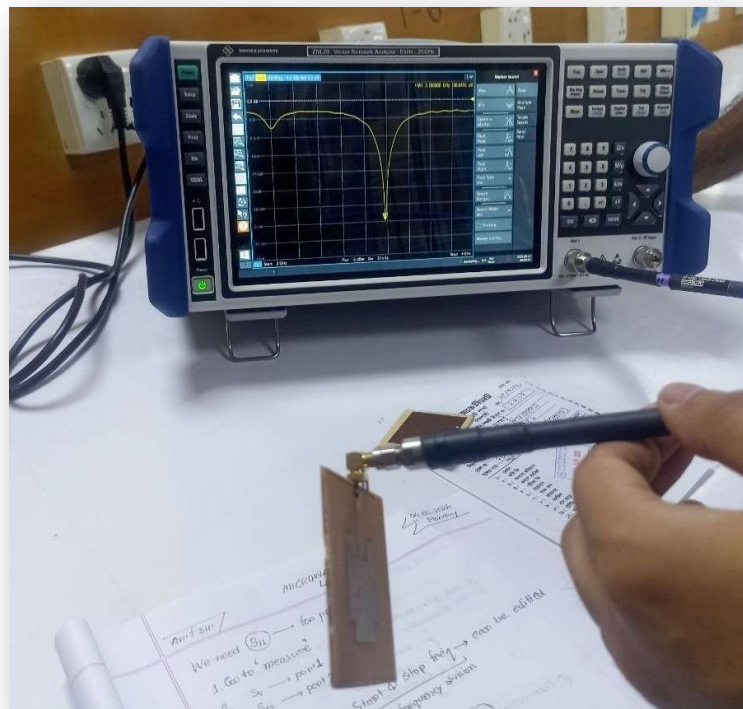


Fig.14: Experimental Setup of the Fabricated Antenna

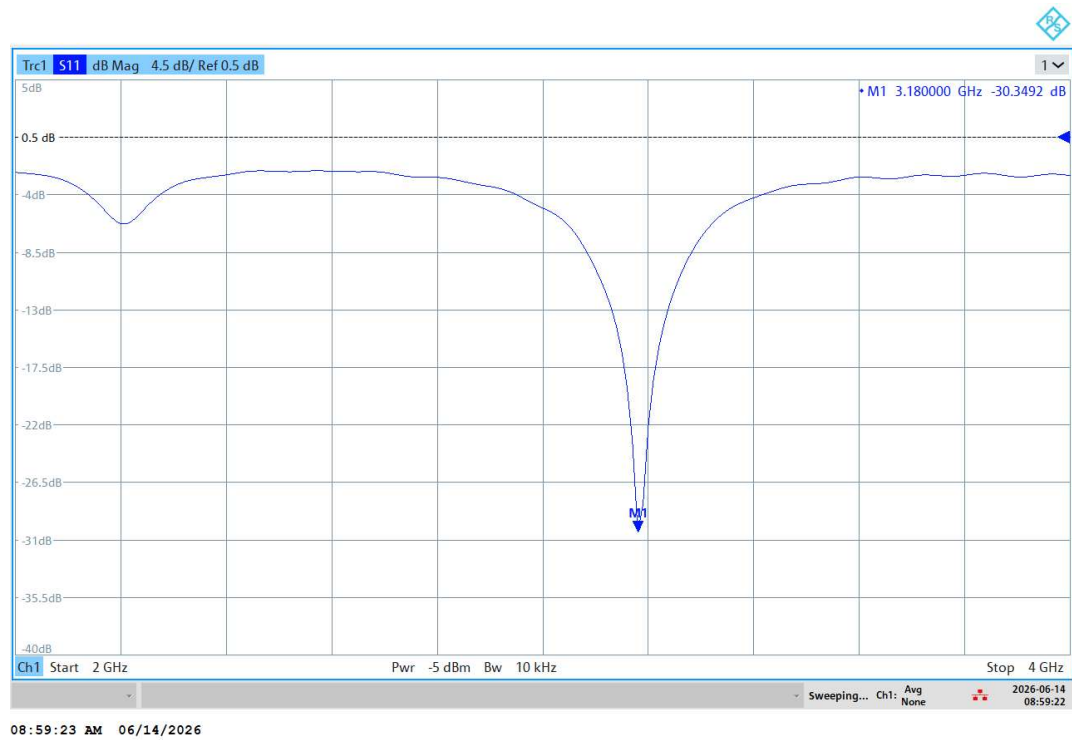


Fig.15: Experimental results of the antenna.

6. Comparison between Simulated and Measured Results

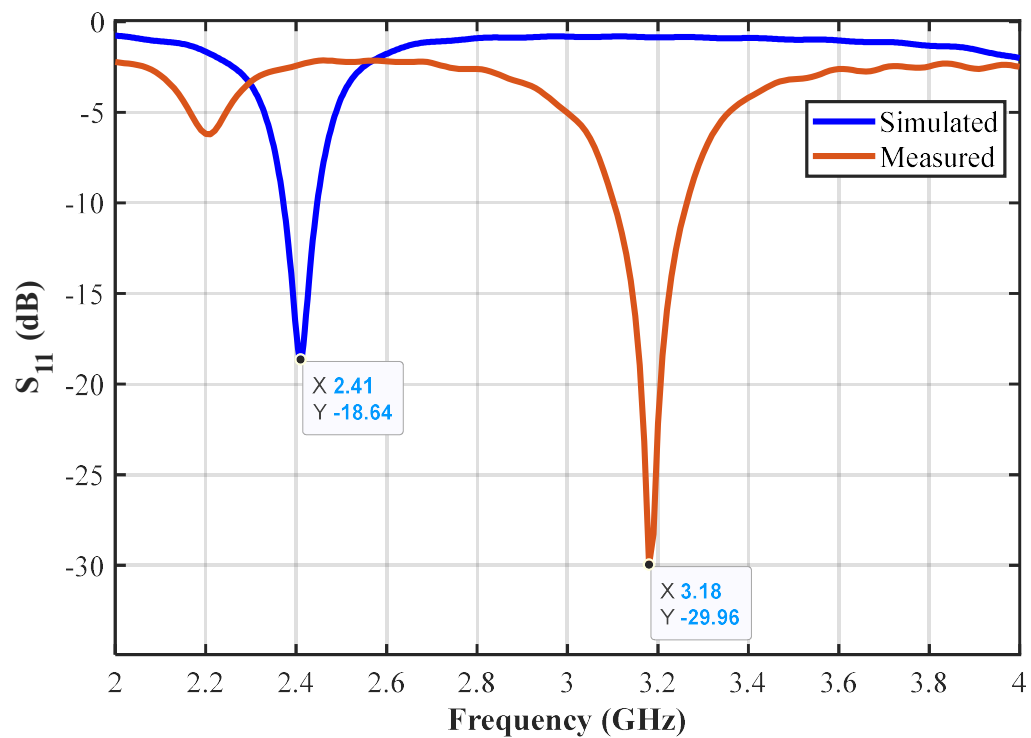


Fig.16: Comparison between simulated and measured results.

Fig.16 shows the comparison between the simulated and measured reflection coefficient S_{11} of the proposed antenna. Both results exhibit a similar resonance behavior, confirming the validity of the antenna design.

The simulated result shows a resonance frequency at approximately **2.41 GHz** with a minimum return loss of **-18.64 dB**, whereas the measured result resonates at approximately **3.18 GHz** with a minimum return loss of **-29.96 dB**. Although both curves demonstrate good impedance matching ($S_{11} < -10$ dB), a noticeable frequency shift is observed between the simulated and measured responses.

The deviation between the simulated and measured results can be attributed to several practical factors:

- **Substrate Material Variation:** The fabricated PCB substrate is not a perfectly homogeneous FR4 material. Therefore, the actual dielectric constant (ϵ_r) may differ from the value used in the simulation.
- **Fabrication Tolerances:** Small deviations in antenna dimensions during PCB manufacturing can significantly affect the resonant frequency and impedance matching.
- **Substrate Thickness Error:** The actual substrate thickness may vary slightly from the designed value, leading to a shift in resonance frequency.
- **SMA Connector and Soldering Effects:** The SMA connector and solder joints introduce additional parasitic inductance and capacitance that are not fully considered in the simulation model.
- **Measurement Uncertainties:** Cable losses, connector mismatches, and environmental reflections during measurement can influence the measured S_{11} response.
- **Material and Conductor Losses:** Copper surface roughness, conductor losses, and substrate inhomogeneity can cause discrepancies between simulated and measured results.
- **Feed Structure Imperfections:** Slight misalignment in the feeding structure or connector attachment can alter the antenna impedance characteristics.
- **Simulation Assumptions:** CST simulations are performed under ideal conditions, whereas practical measurements involve unavoidable real-world imperfections.

Calculation for VSWR and Reflection Coefficient:

The S_{11} values are given in complex form ($\text{Re} + j \cdot \text{Im}$) from the VNA CSV data export. The magnitude in dB is calculated as:

$$S_{11} \text{ (dB)} = 20 \times \log_{10} \sqrt{(\text{Re}^2 + \text{Im}^2)}$$

$$S_{11} = 20 \log_{10} |\Gamma|$$

Simulated Result at $f = 2.41$ GHz:

$$S_{11} = -18.644 \text{ dB}$$

$$|\Gamma| = 10^{\frac{S_{11}}{20}} = 10^{\frac{-18.644}{20}} = 10^{-0.9322} = 0.1168$$

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.1168}{1 - 0.1168} = 1.26$$

Experimental Result at f = 3.18 GHz

$S_{11} = -29.96 \text{ dB}$

$|\Gamma| = 10^{\frac{S_{11}}{20}} = 10^{\frac{-29.96}{20}} = 10^{-1.498} = 0.0318$

$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.0318}{1 - 0.0318} = 1.07$

Table 2: Comparison of Simulated and Measured Results

Parameter	Symbol	Simulated Value	Measured Value
Resonant Frequency	f_r	2.41 GHz	3.18 GHz
Return Loss	S_{11}	-18.644 dB	-29.96 dB
Reflection Coefficient	$ \Gamma $	0.1168	0.0318
VSWR	VSWR	1.26	1.07

7. Addressing Complex Engineering Problems (P’s)

Attributes	Justification
P1 (Depth of knowledge)	The project required the application of electromagnetic theory, antenna design principles, transmission line theory, and microwave engineering concepts. CST Studio Suite was used to model, simulate, and optimize the antenna parameters to achieve resonance near the 2.4 GHz ISM band.
P2 (Range of Conflicting Requirements)	The antenna design involved balancing multiple performance requirements such as compact size, good impedance matching, adequate gain, and acceptable radiation characteristics. Modifications to the patch geometry were performed to achieve the desired performance while maintaining practical fabrication constraints.
P4 (Familiarity of Issues)	The project addressed real-world engineering issues including substrate property variations, fabrication tolerances, SMA connector effects, and measurement uncertainties. Experimental validation using a Vector Network Analyzer (VNA) was carried out to evaluate the practical performance of the fabricated antenna and compare it with simulation results.

8. Conclusion

This project presented the design, simulation, fabrication, and experimental validation of a compact microstrip patch antenna for 2.4 GHz ISM band applications. The antenna was designed on an FR-4 substrate with dimensions of $60\text{ mm} \times 60\text{ mm} \times 1.6\text{ mm}$ and optimized using CST Studio Suite.

The optimized antenna achieved a simulated resonant frequency of 2.41 GHz with a reflection coefficient (S_{11}) of -18.64 dB , indicating good impedance matching. The antenna also exhibited a realized gain of approximately 1.86 dBi and stable radiation characteristics suitable for wireless communication applications. The fabricated antenna was tested using a Vector Network Analyzer (VNA), and the measured results confirmed the resonant behavior of the proposed design.

Although a frequency shift was observed between the simulated and measured results, the discrepancy can be attributed to practical factors such as substrate material variations, fabrication tolerances, SMA connector effects, and measurement uncertainties. Despite these differences, the antenna demonstrated acceptable performance and validated the effectiveness of the proposed design approach.

Therefore, all the objectives of the project were successfully achieved, including antenna design, performance optimization, simulation analysis, fabrication, and experimental verification. The proposed antenna can be considered a suitable candidate for short-range wireless communication systems operating in the ISM band, such as Wi-Fi, IoT devices, and wireless sensor networks.

9. References

- R. M. Saleh, A. K. Jassim, and M. H. Wali, "A Compact Four-Element MIMO Antenna with DNG Metamaterial Decoupling Structure for 2.4 GHz Wi-Fi Applications," *AEM Journal*, vol. 15, no. 1, pp. 24–34, May 2026.
- CST Studio Suite, Dassault Systems, "CST Studio Suite User Documentation," 2024.